

## Raman Kumar Singh

Ph.D. Candidate (Thesis Submitted, July 2026)  
Department of Energy Science and Engineering  
Indian Institute of Technology (IIT) Bombay, Mumbai, India

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🏠 Raman Kumar Singh

### Research Profile

Ph.D. researcher in Energy Science and Engineering (IIT Bombay, thesis submitted July 2026) with six years of combined computational and experimental research experience in **physics-based multiphysics modelling of electrochemical systems**. Developed and validated (COMSOL Multiphysics) models of metal-CO<sub>2</sub> electrochemical cells (Mg-CO<sub>2</sub>, Na-CO<sub>2</sub>, Zn-CO<sub>2</sub>) for simultaneous CO<sub>2</sub> conversion, hydrogen generation, and power production, coupling electrochemical kinetics (Butler-Volmer), ionic and species transport (Nernst-Planck / dilute-solution theory), multiphase flow (Euler-Euler, Darcy, Brinkman, Navier-Stokes), porous-electrode behaviour, species precipitation, anode corrosion, and CO<sub>2</sub> solubility. Skilled at translating electrochemical measurements (EIS, CV, LSV, chronoamperometry) into physics-based, parameter-identifiable models through systematic parametric optimization, sensitivity analysis, and model-experiment validation. Combines this modelling expertise with hands-on experimental work: NASICON solid-electrolyte sol-gel synthesis, custom electrochemical cell design/assembly/testing, and structural/electrochemical characterization (XRD, SEM/EDX). Proficient in Python and MATLAB for data reduction, parameter fitting, and workflow automation. Strong record of independent research leadership, scientific writing, and international presentation: two peer-reviewed publications, one submitted and one further manuscript in preparation, eight international conference presentations across India, Italy, Japan, and New Zealand, a Best Poster Presentation Award, and four years of university teaching experience.

### Education

#### Ph.D. in Energy Science and Engineering

2021 – 2026

Indian Institute of Technology (IIT) Bombay, Mumbai, India

*Thesis submitted July 2026*

Thesis: *Modelling and Analysis of Metal-Carbon Dioxide System for Carbon Utilization and Power Generation*

Supervisors: Prof. Prakash C. Ghosh, Prof. Venkatasailanathan Ramadesigan, & Prof. Amartya Mukhopadhyay

#### M.Sc. in Energy Science and Engineering

2019 – 2021

Indian Institute of Technology (IIT) Bombay, Mumbai, India

Thesis: *A Modeling Approach for Na-Water Based Electrochemical System for CO<sub>2</sub> Capture*

Supervisors: Prof. Prakash C. Ghosh & Prof. Venkatasailanathan Ramadesigan

#### B.Sc.

2015 – 2018

University of Allahabad, Prayagraj, India

### Technical Skills

- **Multiphysics & Process Modelling:** COMSOL Multiphysics – Butler-Volmer electrode kinetics, Nernst-Planck/dilute-solution species transport, Euler-Euler multiphase flow, Darcy/Brinkman/Navier-Stokes flow, porous-electrode theory, species precipitation, anode corrosion, CO<sub>2</sub> solubility, parametric optimization (geometry, flow rate, conductivity, initial conditions); finite element method, finite difference method.
- **Programming & Computing:** Python (NumPy, SciPy, Pandas, Matplotlib); MATLAB; fundamentals of machine learning (regression, classification, model evaluation) via self-study.
- **Electrochemical Characterization:** Electrochemical Impedance Spectroscopy (EIS); Cyclic and Linear Sweep Voltammetry (CV, LSV); chronoamperometry; NASICON solid-electrolyte sol-gel synthesis; X-ray Diffraction (XRD); SEM/EDX.
- **Cell & Stack Hardware:** Custom electrochemical cell design, assembly, and testing; PEM fuel cell and alkaline water electrolyser stack assembly and operation.
- **Research & Communication:** Independent project design and execution from concept to publication; peer-reviewed scientific writing; international conference presentation; university teaching and mentoring; L<sup>A</sup>T<sub>E</sub>X.

## Research Experience

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### Ph.D. Project: Multiphysics Modelling of Metal-CO<sub>2</sub> Electrochemical Systems

July 2021 – 2026

IIT Bombay (Supervisors: Prof. P. C. Ghosh, Prof. V. Ramadesigan, Prof. A. Mukhopadhyay)

- Developed coupled (COMSOL Multiphysics) models of Mg-CO<sub>2</sub>, Na-CO<sub>2</sub>, and Zn-CO<sub>2</sub> electrochemical cells, integrating Butler-Volmer kinetics, Nernst-Planck species transport, Euler-Euler multiphase gas-liquid flow, and Darcy/Brinkman porous-media flow to capture coupled reaction-transport behaviour.
- Modelled electrode-surface and interfacial phenomena – including anode corrosion, species precipitation, and CO<sub>2</sub> solubility – and ran systematic parametric studies across cell geometry, flow rate, electrolyte conductivity, and initial conditions to guide cell design and diagnose transport-limited behaviour.
- Validated the model directly against in-house electrochemical measurements (EIS, CV, LSV, chronoamperometry), closing the loop between simulation and experiment.
- Synthesized NASICON solid electrolytes via a sol-gel route and assembled/operated flow-based metal-CO<sub>2</sub> single cells to generate the experimental datasets used for model calibration and validation.
- Built Python/MATLAB workflows for data reduction, parameter fitting, and post-processing of simulation and experimental datasets.

### Graduate Research Assistant (Fuel Cell Laboratory)

July 2020 – 2026

Department of Energy Science and Engineering, IIT Bombay

- Assembled, operated, and troubleshoot alkaline water electrolyser stacks and PEM fuel cells, applying COMSOL-based multiphysics modelling to interpret electrochemical performance, transport limitations, and gas-phase behaviour.

### M.Sc. Project: Na-Water Electrochemical System for CO<sub>2</sub> Capture

July 2020 – Dec 2021

IIT Bombay (Supervisors: Prof. P. C. Ghosh, Prof. V. Ramadesigan)

- Built and tested a laminar two-phase Na-water electrochemical cell for simultaneous carbon capture and power generation, optimizing reactor geometry, electrode spacing, and CO<sub>2</sub> flow rates through coupled modelling and experimental studies.
- Developed a COMSOL-based transport model of the Na-water cell to interpret the effect of geometric and operating parameters on capture efficiency and power output, validated against bench-scale cell measurements.

## Selected Research Projects

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### Mg-CO<sub>2</sub> Primary Battery – CO<sub>2</sub> Reduction to H<sub>2</sub>

*Electrochimica Acta (2025)*

- Built a 3D transient reactor model coupling reaction kinetics, species transport, multiphase flow, precipitation, and hydrogen evolution to predict CO<sub>2</sub>-conversion performance and identify design levers for improved conversion.

### Na-CO<sub>2</sub> System – Cell Assembly & Model Validation

*NASICON solid electrolyte*

- Assembled a sodium-based CO<sub>2</sub>-conversion cell using a NASICON solid electrolyte; characterized it with EIS, CV, and LSV, analyzed and used these measurements to calibrate and validate a physics-based reactor model.

### Zn-CO<sub>2</sub> Stationary System – CO<sub>2</sub> Utilization & H<sub>2</sub> Production

*Journal of Power Sources (2026)*

- Developed a multiphase-flow reactor model resolving internal gas-liquid transport and porous-media behaviour to isolate mass-transport limitations constraining CO<sub>2</sub>-conversion and H<sub>2</sub>-production performance.

### Zn-CO<sub>2</sub> Flow System – Reactor Sensitivity Study

*Manuscript submitted*

- Ran extensive parametric and sensitivity campaigns on a flow-reactor architecture, quantifying how electrode structure, mass transport, and flow conditions govern CO<sub>2</sub>-conversion selectivity and performance – directly transferable to gas-phase reactor design and outdoor-testing flow-system optimization.

## Selected Publications

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1. **Raman K. Singh**, P. C. Ghosh, and V. Ramadesigan, "Multiphase flow modeling of an aqueous Zn-CO<sub>2</sub> system for CO<sub>2</sub> utilization and hydrogen production," *Journal of Power Sources*, **694**, 241142 (2026). DOI: 10.1016/j.jpowsour.2026.241142.
2. **Raman K. Singh**, P. C. Ghosh, and V. Ramadesigan, "Mathematical modeling and simulation of Mg-CO<sub>2</sub> primary battery for hydrogen generation and CO<sub>2</sub> utilization," *Electrochimica Acta*, **531**, 146333 (2025). DOI: 10.1016/j.electacta.2025.146333.
3. **Raman K. Singh**, P. C. Ghosh, and V. Ramadesigan, "A flow model for simultaneous CO<sub>2</sub> conversion, H<sub>2</sub> production, and power generation in a Zn-CO<sub>2</sub> electrochemical system," manuscript submitted.
4. **Raman K. Singh**, D. Saha, S. Karmakar, P. C. Ghosh, A. Mukhopadhyay, and V. Ramadesigan, "Modeling and experimental investigation of a Na-CO<sub>2</sub> system for hydrogen and power generation," manuscript in preparation.

## Conference Presentations & Awards

- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Model-based investigation of Zn-CO<sub>2</sub> electrochemical systems for CO<sub>2</sub> reduction to H<sub>2</sub> and electricity production. *SustainX – ACE 2025 & 4th CO2India Meet*, IIT Madras, Chennai, India. **[Best Poster Presentation Award]**
- **Raman K. Singh**, D. Saha, S. Karmakar, P. C. Ghosh, A. Mukhopadhyay & V. Ramadesigan. Multifunctional Na-CO<sub>2</sub> electrochemical system: experimental validation and modeling for sustainable hydrogen and NaHCO<sub>3</sub> co-production. *ACEPS 13*, 2026, Bengaluru, India. (Poster)
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Flow-based Zn-CO<sub>2</sub> battery for simultaneous CO<sub>2</sub> mitigation, power generation, and hydrogen production. *ICNEA 2025*, Tokyo, Japan.
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. A combined chemical, electrochemical, and multiphase flow model of Zn-CO<sub>2</sub> system for CO<sub>2</sub> utilization in H<sub>2</sub> and electricity. *ICOSSE'25*, 2025, Auckland, New Zealand. (Poster)
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Study on the effects of bubble diameter and reaction rate kinetics in an Mg-CO<sub>2</sub> electrochemical system. *EIHE 2025*, Mumbai, India. (Poster)
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. A mathematical modeling approach to utilize CO<sub>2</sub> into hydrogen and electricity via Na-CO<sub>2</sub> electrochemical system. *CACEE 2024*, TIFR, Mumbai, India. (Poster)
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Modeling and numerical study of Mg-CO<sub>2</sub> electrochemical system for CO<sub>2</sub> utilization with electricity and hydrogen production. *37th Topical Meeting of the International Society of Electrochemistry (ISE)*, 2024, Stresa, Italy. (Poster)
- **Raman K. Singh**, P. C. Ghosh, & V. Ramadesigan. Electrical field effect on CO<sub>2</sub> absorption and chemisorption in a rectangular bubble column by COMSOL. *Proceedings of the COMSOL Conference 2023*, Bengaluru, India.

## Course Projects

- **Diffusion measurement using electrochemical method** [Jan'21 – May'21]
  - Estimation of diffusion coefficient by linear sweep voltammetry for ionic species.
- **Cathode materials and catalyst effect on the performance of Na-CO<sub>2</sub> batteries** [Jan'21 – May'21]
  - Conducted a literature review of Na-CO<sub>2</sub> batteries, focusing on capacity, coulombic efficiency, and cyclic stability in relation to variations in cathode catalysts.
- **Mathematical modeling of multiple wake effect in wind farms** [July'20 – Nov'20]
  - Calculated the velocity profile from downstream to upstream turbines for a specific wind farm configuration, accounting for wake effects based on turbine specifications and farm area.
- **Assessment of behavioural impact on environment and energy** [July'20 – Nov'20]
  - Analyzed the environmental impact of daily activities through carbon footprint assessment.
- **Simulation of building for minimum cooling energy requirement** [July'19 – Nov'19]

- Simulated a building (Shunya House) and calculated the minimum energy required to cool the building to a target temperature by varying wall material, glass type, orientation, and building location.
- **Solid Oxide Fuel Cell Microstructure and Properties**  
(*M.Sc. Seminar / Guide: Prof. Dayadeep Moder*) *[Jan'20 – May'20]*
  - Analyzed the three-phase boundary and contact area per unit volume, hydraulic radius, and effective conductivity, focusing on changes in material properties such as particle size of electron- and ion-conducting materials, porosity, pore-former radius, and contact angle.

## Key Courses

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|-----------------------------------------------|---------------------------------------------------|
| • Energy System Modelling & Analysis          | • Electrochemical Materials Science               |
| • Energy Resources, Economics and Environment | • Numerical Methods for Environmental Systems     |
| • Methods in Analytical Techniques            | • Introduction to Transport Phenomena             |
| • Electrochemical Reaction Engineering        | • Principles and Applications of Hydrogen Storage |

## Teaching Experience

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### Teaching Assistant

July 2022 – Dec 2025

*Department of Energy Science and Engineering, IIT Bombay*

- **EN604 Fuel Cells** (2023): Delivered tutorial sessions and evaluated assignments on fuel cell electrochemistry, transport phenomena, and stack design fundamentals for postgraduate students.
- **EN624 Conservation of Energy in Buildings** (2022, 2023 – two semesters): Assisted with course delivery, problem sets, and student project evaluation on building energy audits and efficiency measures.
- **Energy Laboratory** (2022): Supervised undergraduate laboratory sessions covering fundamental energy conversion and measurement experiments.
- **Non-Conventional Energy Systems Laboratory** (2022–2025): Supervised hands-on laboratory sessions on solar, wind, fuel cell, and electrolyser experiments, mentoring 100+ undergraduate and postgraduate students in total across all TA roles.

## Leadership & Service

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- Organizing Committee Member, 9th International Conference on Advances in Energy Research (ICAER), 2023.
- Core Member, Energy Day event, 2021.
- Mentor, Institute Research Student Camping Programme (IRSCP), IIT Bombay, 2022–2023.